

Alexander Ruse

📍 Boston, MA ✉ aruse@umass.edu ☎ (858)-342-8284

Education

University of Massachusetts, Amherst

Candidate, May 2026

Masters of Science in Applied Mathematics

University of Massachusetts, Amherst

2022 - 2024

Bachelor of Science in Mathematics

Relevant Coursework and Skills

Relevant Coursework: Data Structures, Measure Theory, Mathematical Machine Learning, Bayesian Modeling, Reinforcement Learning, Stochastic Processes, PDEs, Longitudinal Data Analysis, Causal Inference

Programming Languages: Python, R, Java, SQL, MATLAB, SAS, LaTeX

Machine Learning & Statistics: torch, scikit-learn, brms, Gurobi, RStan, statsmodels, econml

MLOps & Tools: Databricks, MLflow, Docker, Git, Snowflake

Work Experience and Relevant Projects

Causal Modeling for Insurance Optimization at Liberty Mutual

June - August 2025

- Analyzed patterns of missingness and selection bias in insurance data using nonparametric tests (Kolmogorov–Smirnov, Mann–Whitney U) to inform causal model assumptions
- Designed and implemented a framework for modeling using CausalDML to improve out-of-sample prediction under treatment assignment bias
- Built a production-grade **MLOps** pipeline on **Databricks** to optimize default policy recommendations, integrating LightGBM, EconML Causal Forests, and MLflow for reproducible experimentation and version control

Anticipatory Prioritized Emptying for Dynamic Resource Allocation

March - May 2025

- Developed the **Anticipatory Prioritized Emptying (APE)** algorithm to optimize container evacuation using Kalman-filtered forecasts and a reward-aligned priority function; formulated a custom MDP with Gaussian fill-rate dynamics and overflow penalties
- Integrated APE into an Actor-Critic agent via augmented state vectors; in 500-episode simulations, **ForecastAPE** improved average reward by **214%** and outperformed both Actor-Critic and rule-based agents in robustness and sample efficiency
- Engineered full simulation stack using Python and Gym: parameterized stochastic environments with container-specific drift, performed reward shaping, hyperparameter tuning, and constructed reproducible training pipelines with logging, dataclasses, and ablation tracking

Bayesian Hierarchical Modeling for COVID Hospitalization

November-December 2024

- Developed a Bayesian hierarchical logistic regression model to predict hospitalization risk from CDC COVID surveillance data, incorporating **state-level random effects** to capture geographic variability in under-reporting.
- Improved model performance over a non-hierarchical baseline by **104.9 elpd points (SE = 15.8)**, demonstrating significantly better predictive accuracy and generalization to low-sample states.
- Used brms in R to fit models across imputed datasets, addressing missingness in age and race; visualized results via predictive probability maps and credible intervals for public health resource allocation.

Score-Based Generative Modeling using HJB Regularizers

May - July 2023

- Implemented potential implicit score-matching algorithms in PyTorch to approximate high-dimensional solutions to mean-field games (MFGs) and HJB PDEs
- Developed and trained deep neural networks using a potential-based score-matching loss with Hamilton-Jacobi-Bellman regularization, achieving visually improved sample fidelity and 30,000-epoch stability in high-dimensional synthetic and image datasets (MNIST, CIFAR-10)